



## Otto DIY build your own robot



HP Robots Otto

[VIEW IN BROWSER](#)

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### Summary

An interactive robot that anyone can make, program with Arduino or Blockly software, remote control and expandable.



5.58 hrs



2 pcs



0.28 mm



0.40 mm



PLA



92 g



Prusa MINI /  
MINI+

[Hobby & Makers](#) > [RC & Robotics](#)

Tags: [toy](#) [robot](#) [steam](#) [diy](#) [stem](#) [biped](#) [arduino](#) [iot](#)  
[ottodiy](#)

### An interactive robot that anyone can make!

Otto walks, dances, makes sounds, avoids obstacles, you can code on your own and even customize, it is completely open source, Arduino compatible and 3D printable.

<https://www.youtube.com/watch?v=np4iqqFFoc4>

### Features

- Programmable with Otto Blockly or Arduino IDE

- Walks & dances
- Avoids obstacles
- Makes emotional sounds and melodies

more details in <https://www.ottodiy.com/>

## Print instructions

Otto is very well designed for 3D printing, so won't give you trouble if you follow these common parameters:

**FDM 3D printer, PLA is enough.**

**No supports or rafts.**

**Resolution: 0.2mm**

**Fill density 15%**

All parts in total use 100gr aprox. equivalent to 35m of regular PLA 1.75mm thickness, it should take around 4 hours to 3D print a full set of parts for one Otto even less with optimized settings depending on your printer and slicer.

## List of Parts

- Otto Nano Microcontroller + I/O board
- USB cable
- Rechargeable 16340 battery booster **do NOT use Alkaline not enough power or use**
- Toggle or 8x8 push switch depending on the body your print
- ⚙ 4 x micro servo motors with set of screws.
- Ultrasonic sensor **HC-SR04**
- Buzzer passive Ø12mm
- 8x DuPont female to female jumper wires
- Small Phillips screwdriver magnetized
- 3D Printed head
- 3D Printed body
- 3D Printed legs (2)
- 3D Printed feet (2)

## How to build video

<https://www.youtube.com/watch?v=h3jog4-qTQY>

- Expand into a [Humanoid robot](#) or [Smart robot](#)
- or simply decorate with these [robot face accessories](#)

We invested lots of time and resources to provide open-source code, software, and hardware, please support this project by just **following us**,

giving us a ♥ **Like and share** and you are **welcome to be a part of this friendly community** of robot builders, teachers, and makers.

**Join our HP Robots Otto: Remix Challenge [click here](#)**

## This remix is based on



**Otto DIY is a 3D printable open source robot**

## Model files



**ottodiyheadv13\_nanoshield.stl**

☐ Head to use when having the Arduino Nano + I/O Shield



**ottodiybodyv11\_booster-square-switch.stl**

☐ Body for booster + charger battery module (compatible with toggle switch)



**ottodiylegv13.stl**

☐ Print x2



**ottodiyfootrv13.stl**

☐ Foot Right



**ottodiyfootlv13.stl**

☐ Foot Left



**power\_switch\_button.stl**

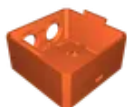
☐ Button cap for micro switch (optional)



### ottodiyhead\_esp8266.stl

☐ Alternative when having NodeMCU ESP8266 + I/O Shield

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### ottodiyhead\_ninjaboard.stl

☐ Alternative when having diymore PCB

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### ottodiybody\_9vbattery.stl

☐ Alternative when having the 9V Lithium rechargeable batteries

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### ottodiybody\_booster-toggle-switch.stl

☐ Alternative when having toggle switch

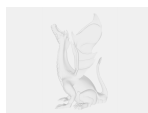
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### otto-diy-cad-step.step

☐ CAD export neutral

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### ottodiy-v129.f3z

☐ CAD source Fusion 360 (version differs from STL)

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## Print files



### ottodiyhead\_028mm\_pla\_mini\_1h28m.gcode

⚙️ PLA ⚙️ 0.40 mm ⚙️ 0.28 mm ⌚ 1.46 hrs ⚖️ 26 g 🖨️ Prusa MINI / MINI+

☐ Head only

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### ottodiybodylegsfeet\_028mm\_pla\_mini\_4h7m.gcode

⚙️ PLA ⚙️ 0.40 mm ⚙️ 0.28 mm ⌚ 4.12 hrs ⚖️ 66 g 🖨️ Prusa MINI / MINI+

☐ Body legs and feet together

## Other files



**ottodiy\_manualsparkfun\_shieldnano.pdf**

📄 this one is specifically for the standard kit

[Find source .stl files on Thingiverse.com](#)

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